

ECON 695 Problem Set 4

October 10, 2025

Due: 11:59PM Central Time on Friday, 10/24/2025.

Note: I provide a Jupyter notebook (**ssp.ipynb**) to guide the coding exercise (question 2).

Reminder 1: If you use AI assistance, make sure you have an AI disclaimer that indicates where/how you use AI tools. Grader will take 2 points off if she recognizes AI-generated code but does not see a disclaimer.

Reminder 2: Some students forgot to upload answers to the theory part of the problem set (e.g., question 1). Please make sure to upload your answers completely. Answer to Question 1 could be a photocopy of your handwritten solution. For Question 2, upload your code **AND** the final output as a HTML/PDF.

Question 1: Average Treatment Effects

Suppose we have a random sample of size N of which $N_1 < N$ were randomly assigned to the treatment group $D_i = 1$, and for these individuals we observe $Y_i = Y_i(1)$, while the rest $N_2 = N - N_1$ were assigned to the control group $D_i = 0$, for whom we observe $Y_i = Y_i(0)$.

1. Show that under random assignment of treatment, the ATE ($\alpha = E[Y_i(1) - Y_i(0)]$) is equal to another causal effect of interest, which is Average Treatment on the Treated $ATT := E[Y_i(1) - Y_i(0)|D_i = 1]$. Generally speaking, how would you interpret ATE versus ATT?
2. Show:

$$\begin{aligned} E[Y_i D_i] &= E[Y_i(1)|D_i = 1] \times Pr(D_i = 1) \\ E[Y_i(1 - D_i)] &= E[Y_i(0)|D_i = 0] \times Pr(D_i = 0) \end{aligned}$$

3. Show that under random assignment of treatment,

$$\hat{\alpha} = \frac{\sum_{i=1}^N D_i Y_i}{\sum_i D_i} - \frac{\sum_{i=1}^N (1 - D_i) Y_i}{\sum_i (1 - D_i)}$$

is a consistent estimator for $\alpha = E[Y_i(1)] - E[Y_i(0)]$.

Question 2: Self Sufficiency Program (see ssp.ipynb)

Background from Card and Hyslop (2005):

- In the Self Sufficiency Project (SSP), members of a randomly assigned treatment group could receive a subsidy for full-time work. The subsidy was available for 3 years, but only to people who began working full time within 12 months of random assignment.
- SSP provides two incentives: (1) find a job and leave welfare within a year, and (2) choose work over welfare in the longer term.
- This question is designed to apply the IV-LATE framework to estimate the causal effects of SSP on leaving welfare, and characterize the compliers who would not have found a full-time job in the absence of the SSP subsidy. The data set welfare.csv contains 5,480 observations for people in the SSP experiment. The data set has the following variables:

- **treatment** = 1 if in treatment group
 - **ft15, ft20, ft24, ft48** = set of dummies if working full time at months 15, 20, 24, and 48 after random assignment
 - **welfare15...welfare48** = set of dummies if receiving welfare at months 15, 20, 24, and 48 after random assignment
 - Other characteristics:
 - * **imm** = 1 if immigrant
 - * **hsgrad** = 1 if high school graduate (or higher) education
 - * **agelt25** = 1 if age is 25 or less
 - * **age35p** = 1 if age is 35 or more
 - * **working_at_baseline** = 1 if person was working when assigned to treatment/control group
 - * **anykidsu6** = 1 if any children under age 6
 - * **nevermarried** = 1 if never married (all cases are single parents)
1. (First stage) Estimate first stage models for the probability of working FT in months 15, 20, 24, 48, using treatment as the instrumental variable.

$$FT_i(t) = \pi_0 + \pi_1 \text{treatment}_i + \epsilon_i, \text{ for } t = 15, 20, 24, 28$$

What does π_0 represent? What does π_1 represent?

2. (Reduced form) Estimate reduced form models for the probability of being on welfare in months 15, 20, 24, 48, using treatment as the instrumental variable.

$$\text{Welfare}_i(t) = \delta_0 + \delta_1 \text{treatment}_i + \nu_i, \text{ for } t = 15, 20, 24, 28$$

where $\text{Welfare}_i(t) = 1$ if person i is on welfare t months since random assignment.

3. (2SLS) Estimate 2SLS models of the equation:

$$\text{Welfare}_i(t) = \beta_0 + \beta_1 FT_i(t) + u_i, \text{ for } t = 15, 20, 24, 28$$

using treatment as an instrument for $FT_i(t)$. Verify that the 2SLS estimate in each case is the ratio of the reduced form and first stage coefficients. Hint: using `linearmodel IV2SLS`.

4. (Characteristics of Compliers) Find the mean characteristics of the compliers in month $t = 15$: i.e., means of the variables **imm**, **hsgrad**, **agelt25**, **age35p**, **working_at_baseline**, **anykidsu6**, and **nevermarried**. Compare these to the characteristics of always takers and never takers in month 15. Hint: define outcome $Y_i = X_i \times FT_i(15)$.